



DevOps

Dr. John-Paul Ore

Welcome aboard!

I Hope you have a productive and positive learning experience 😊

About me...

- Joined NC State in 2019.
- I've lived in Nebraska, Chicago, Seattle, and Raleigh.
- Research is on Software for Mobile Robots, LLMs + Transformers.
- I like gardening, cycling, plants.

Today's Learning Objectives

“Administrivia”

- Understand course structure, high-level cadence.
- Familiar with the Syllabus (grading, late work, attendance policy).
- How to seek help in CSC-519.

DevOps

- Software development review: “Git Flow.”
- DevOps at a high level – what is it?

DevOps CSC-519 : 3 Populations

This class has 2 sections:

- CSC-519-010 (grad students)
- CSC-495 “Special Topics” (high-level undergrads)
- **FYI Panopto: you’re being recorded during class.**

Course Website

<https://github.ncsu.edu/CSC-519/course> is the primary way to find resources for the course:

- course calendar
- [the syllabus](#)
- slides (posted before class)
- links to workshops

(Tour of website)

Workshops

- Linked from the course website.
- Assigned on Wednesdays, due the following Tuesday night.
- Usually, means submitting a Google Form (possibly with screenshots + code)

Exams

- Midterm & Final.
- Paper exams. Pen / Pencil. In class.

Class Project: DevOps pipeline

- 2nd Half of Semester
- Teams of 1-3 people.
- ~5 weeks.
- Using github.ncsu.edu.

519 Weekly Cadence (*1st Half of Semester*)

MONDAYS

- Lectures (& sometimes a head start on Workshops).

WEDNESDAYS

- Workshops / Project Elements
 - live help during class time 3:00-4:15pm

Due the follow TUESDAY MIDNIGHT AOE ([Anywhere on Earth time](#))

- Workshop / Project due from previous week.

DevOps Syllabus

- [Link](#) ← You are responsible for reading and understanding this.
- Attendance Policy ← Randomly taken. **3 Absences** without penalty.
- Late Assignment Policy ← *No late assignments without prior approval.*
- AI Policy ← Encouraged. *Always acknowledge when AI is used.*

Help Seeking

- [Q&A “EdStem Discussion”](#) ← you should have received an invite via email. If not, please contact the TAs
- Professor’s Office Hours (Mondays 1:15-2:15pm)

Help Seeking: DevOps TAs

Kaushik

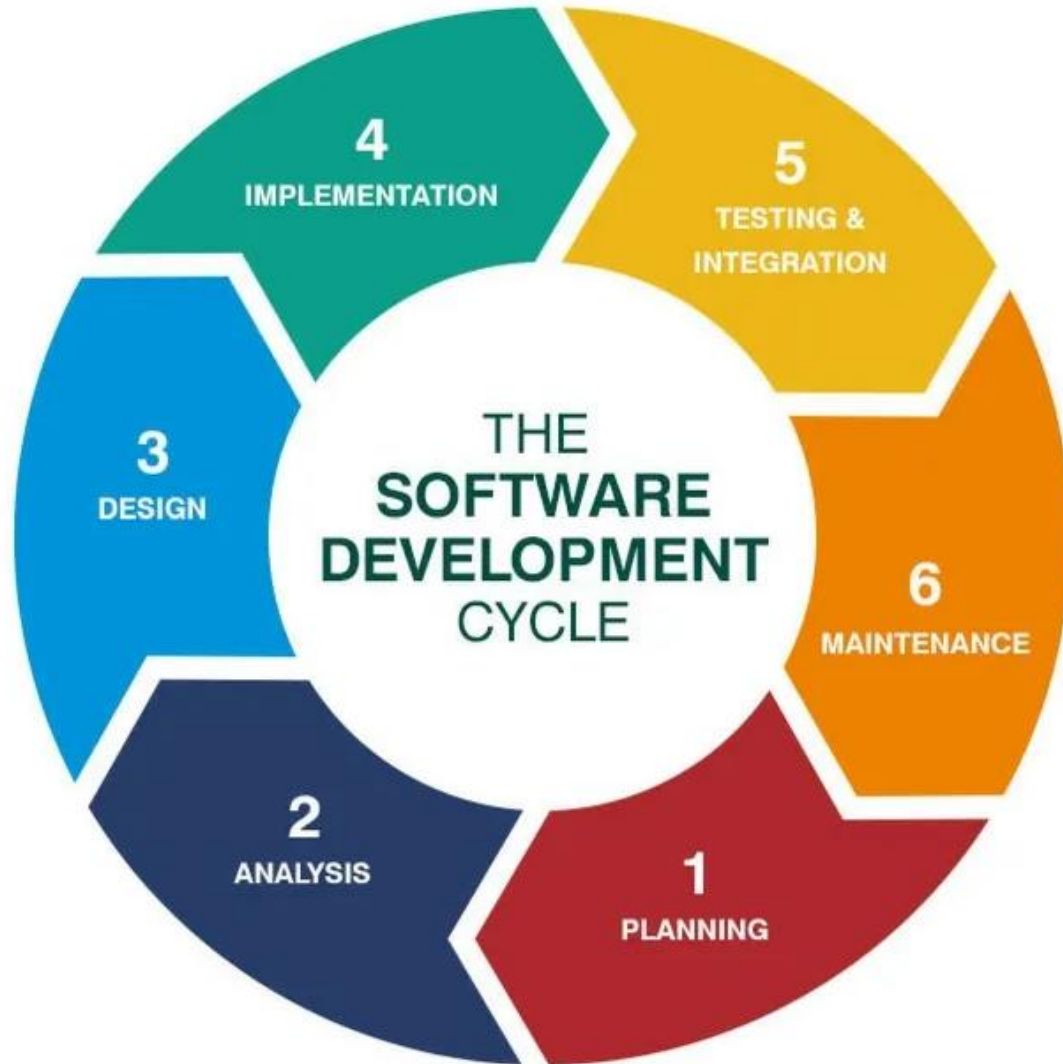
- spillal2@ncsu.edu

Liwen

- ldu2@ncsu.edu

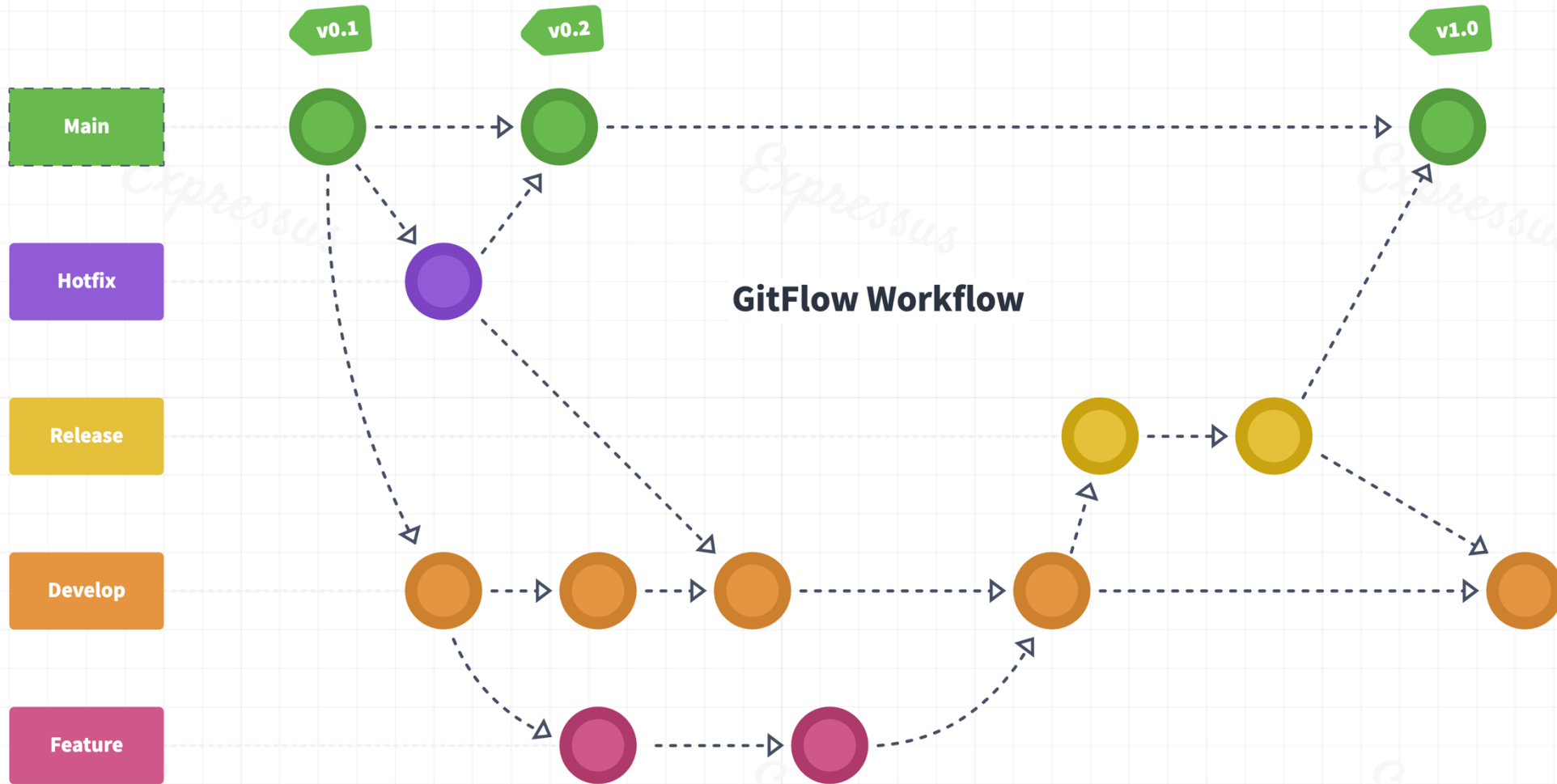
“Git Flow” Dev Process

Software Development as a “Lifecycle”



A Git-based Development Workflow a.k.a “Git Flow”

Important
slide



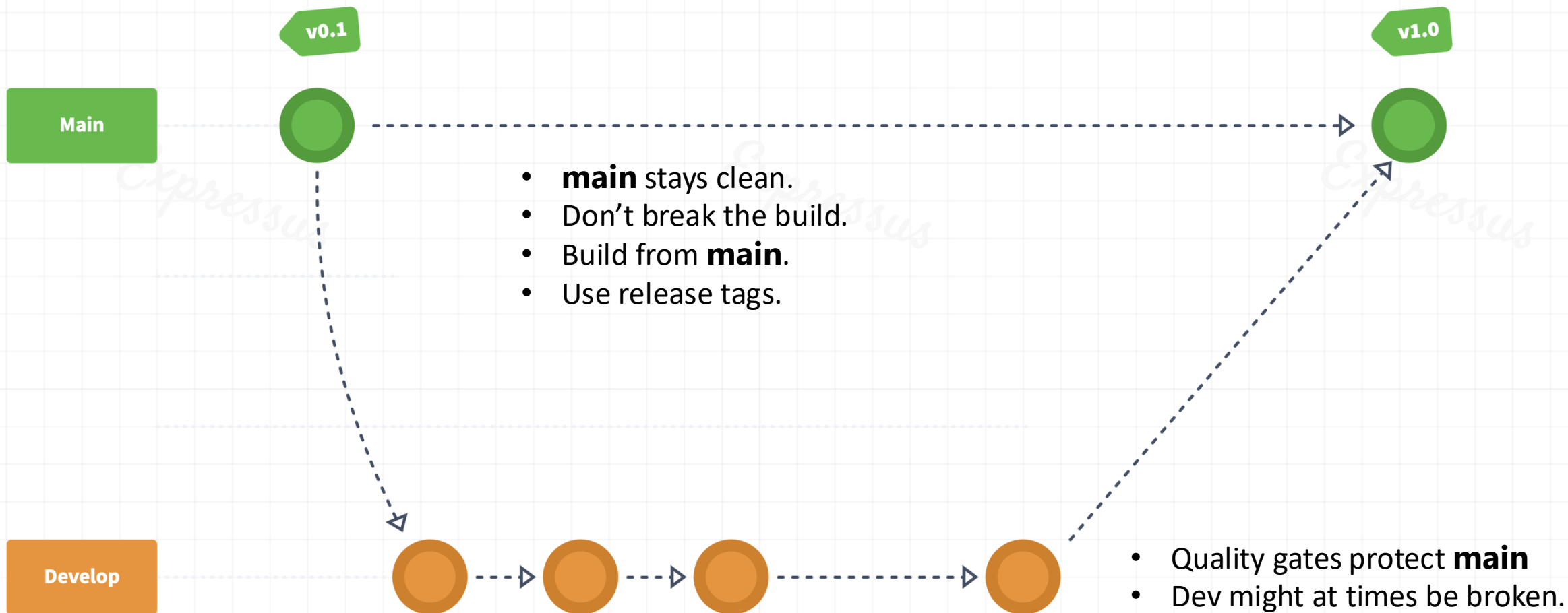
A Main branch...with tags



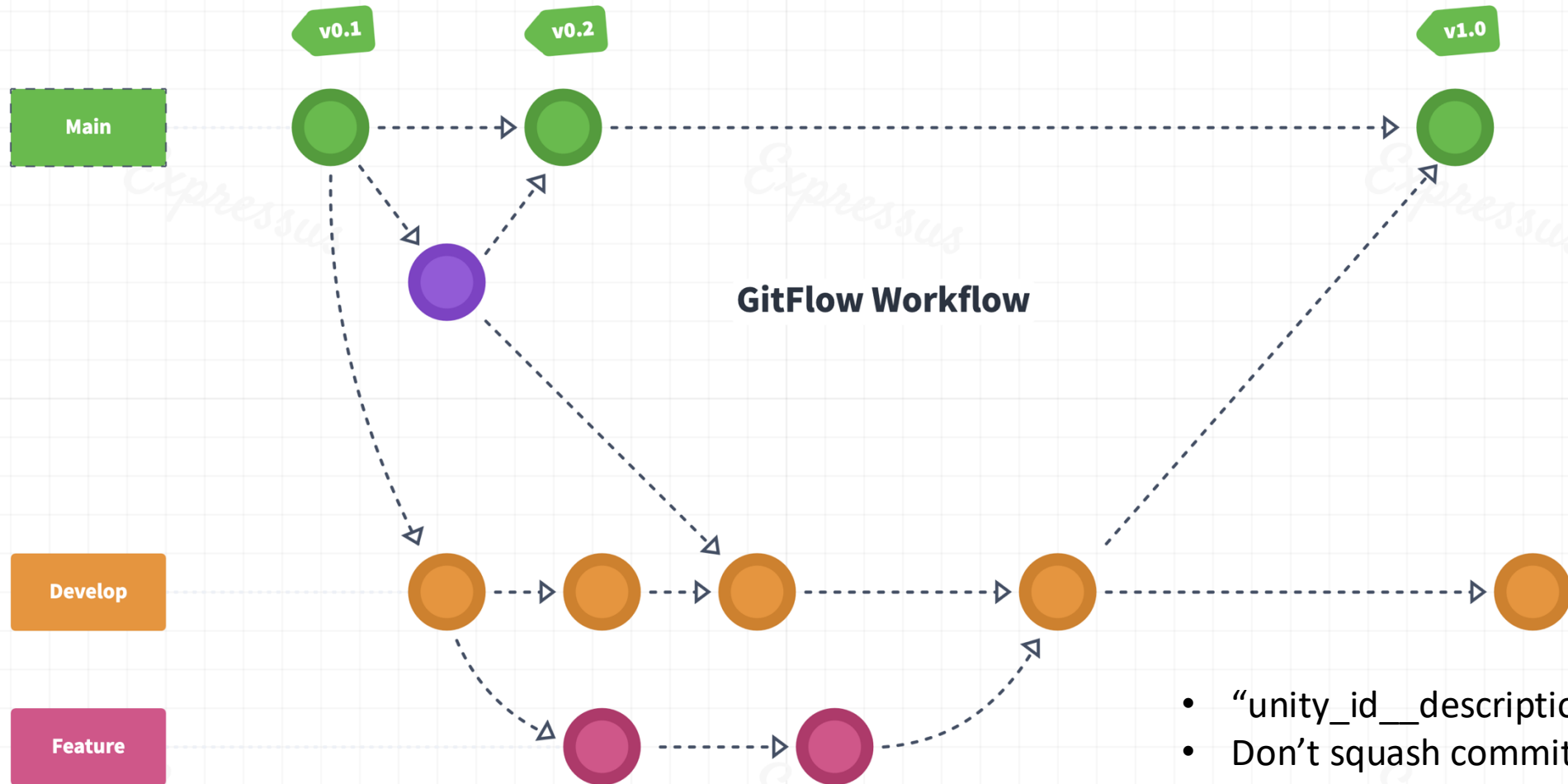
Hotfix



Main & Dev



Feature Branches



What is DevOps?

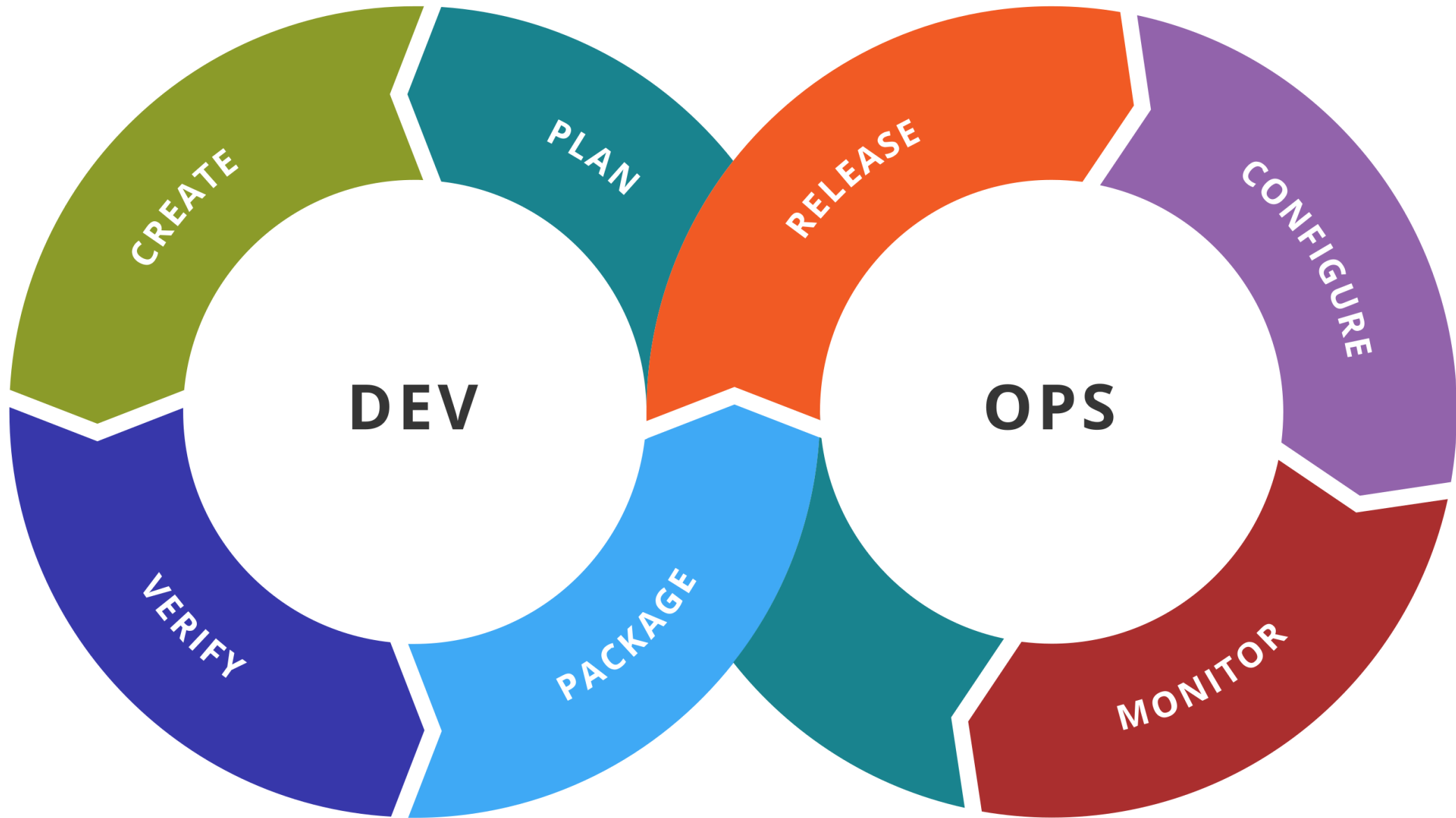
What is DevOps? It's about Perspective

- *“DevOps’ is a set of practices designed to help teams build, test, and release software faster and more reliably.”* (src: <https://gearset.com/blog/salesforce-devops-what-is-it/>)

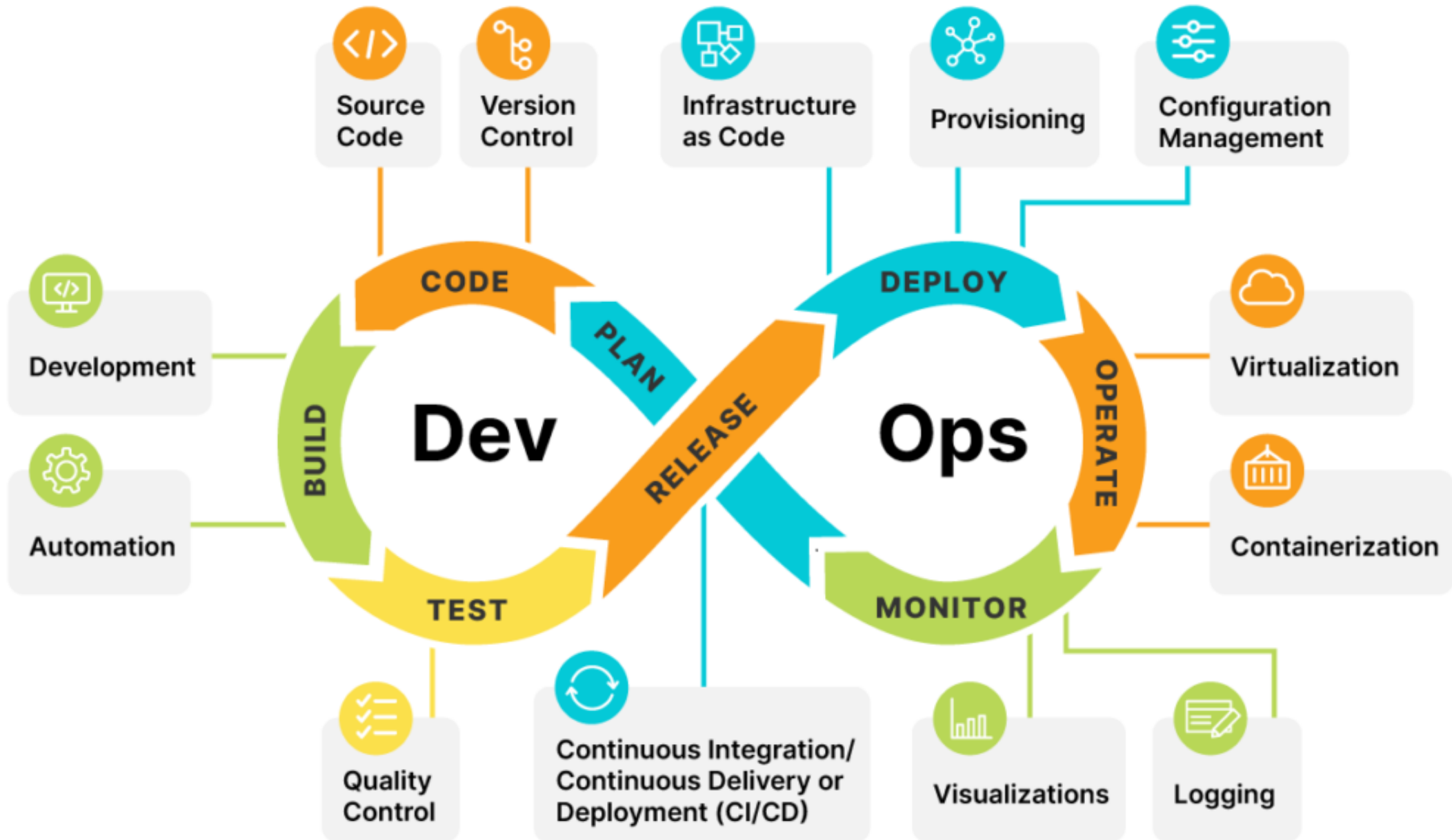
Development

Operations

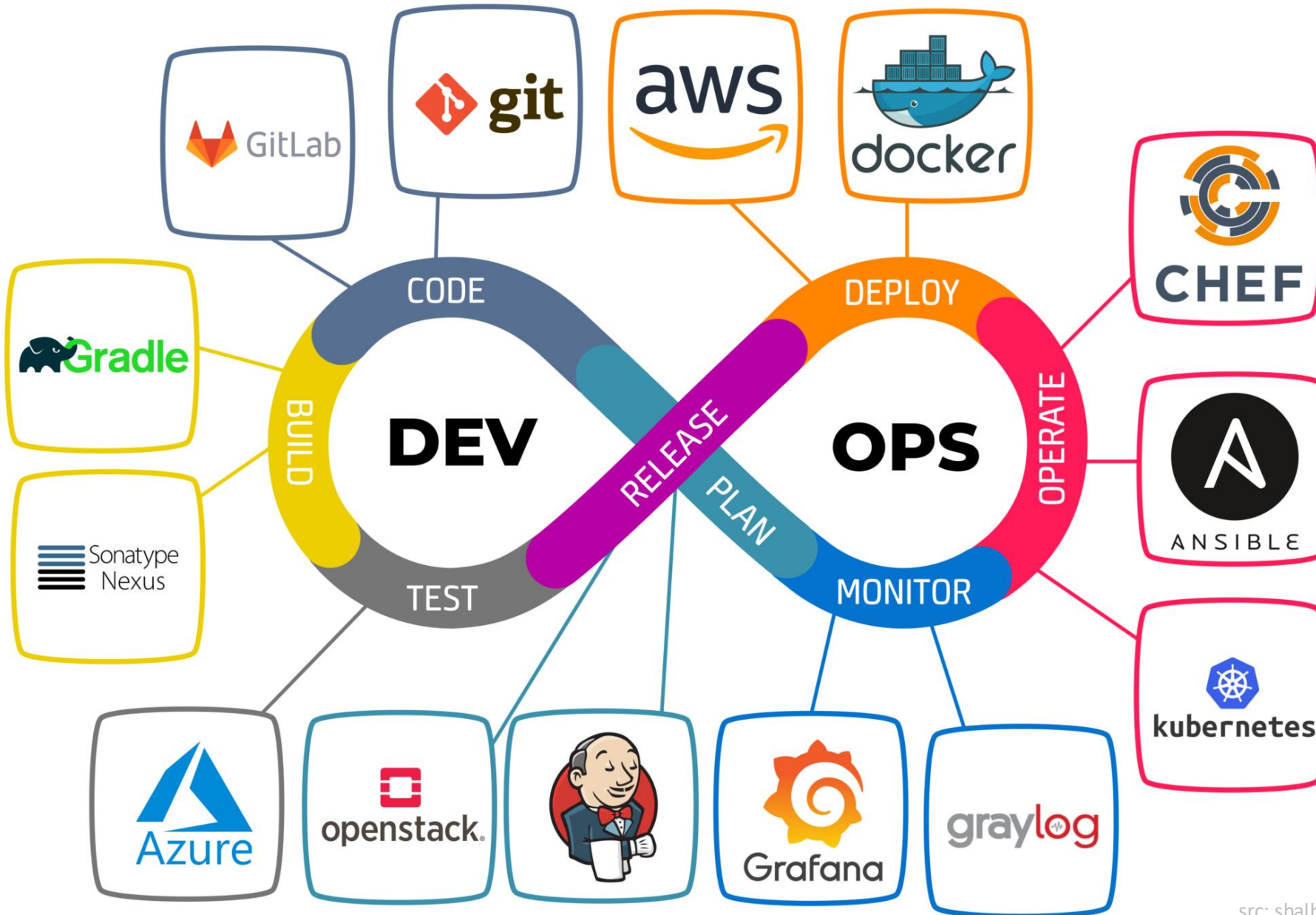
What is DevOps?



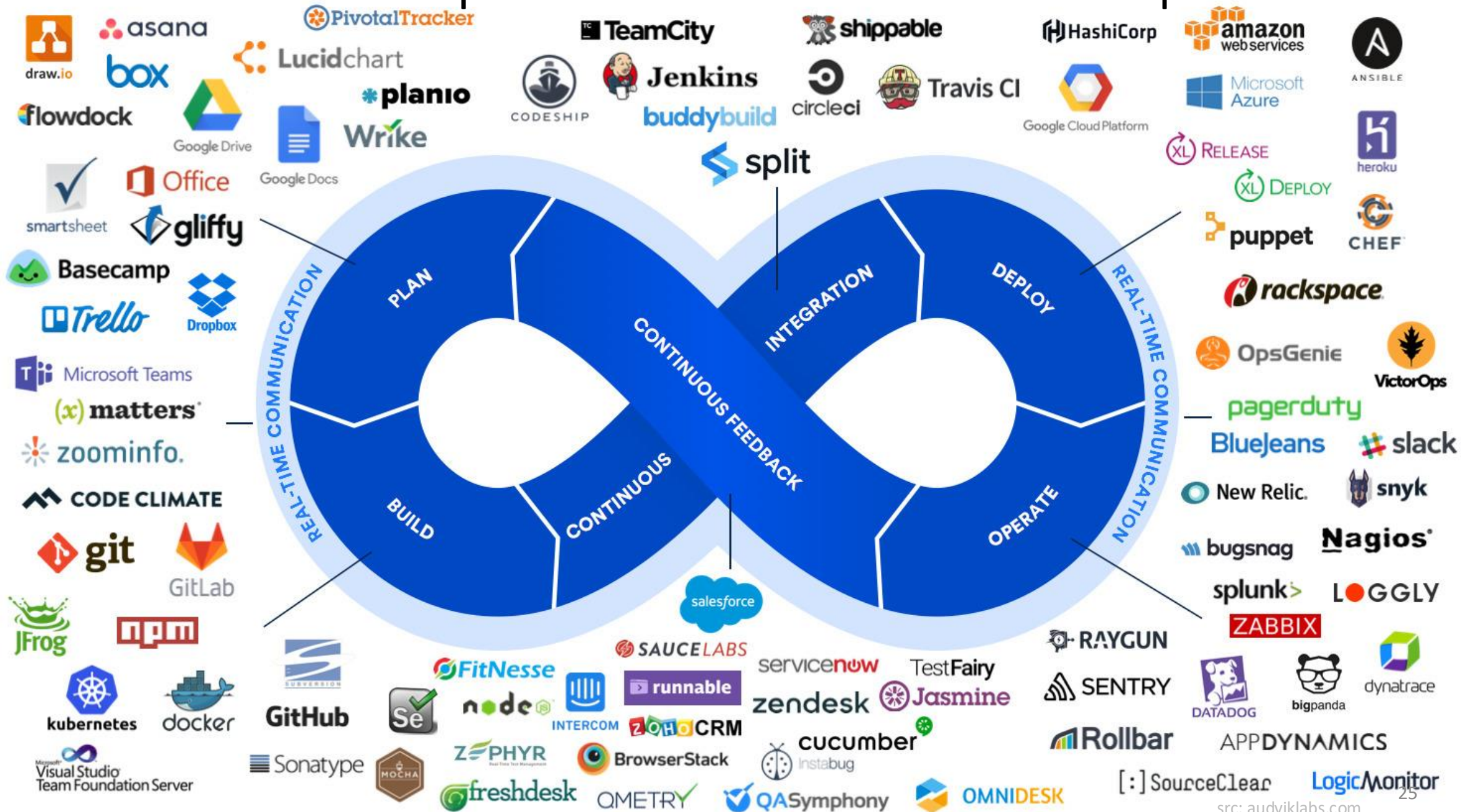
What is DevOps?



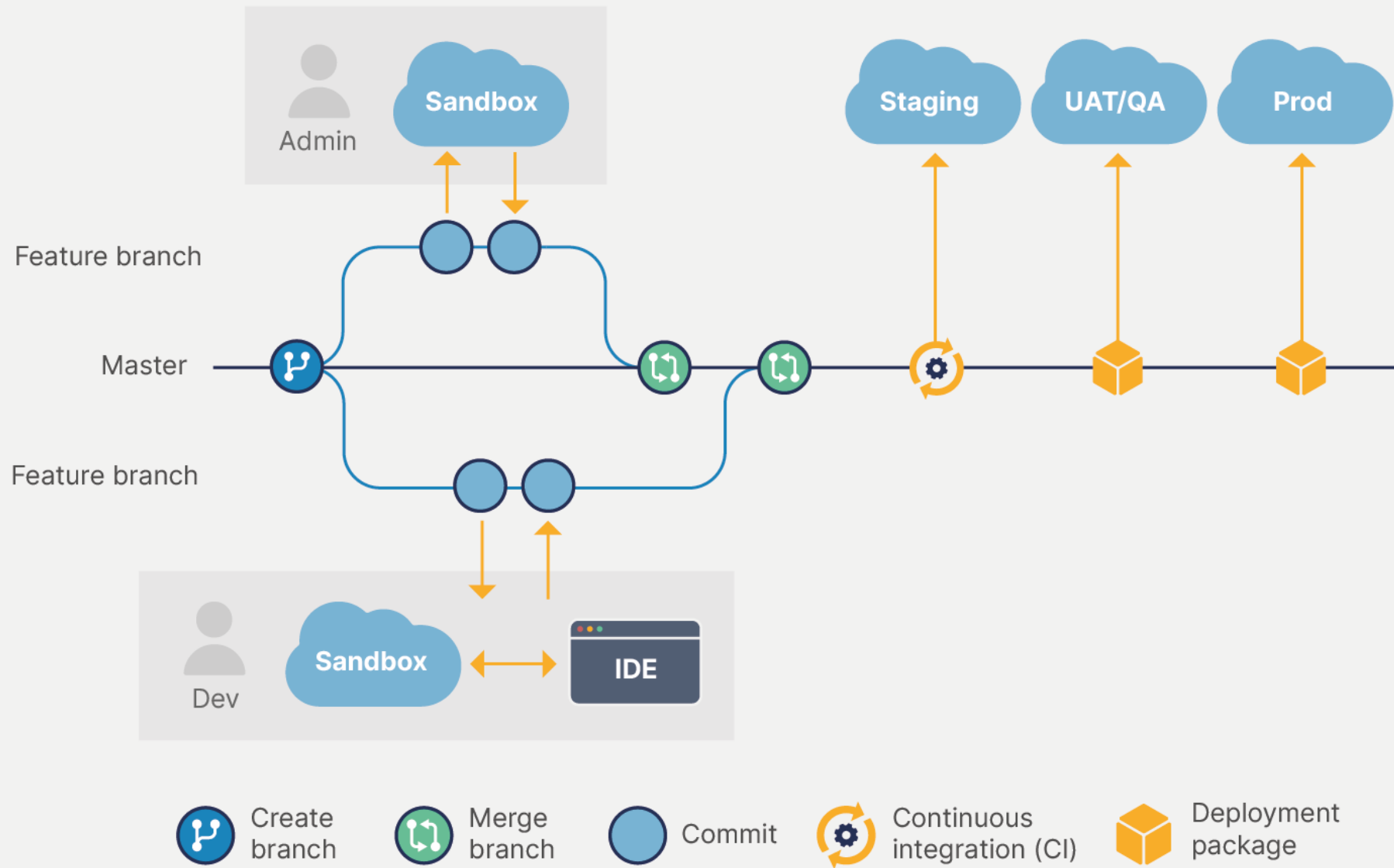
What is DevOps?



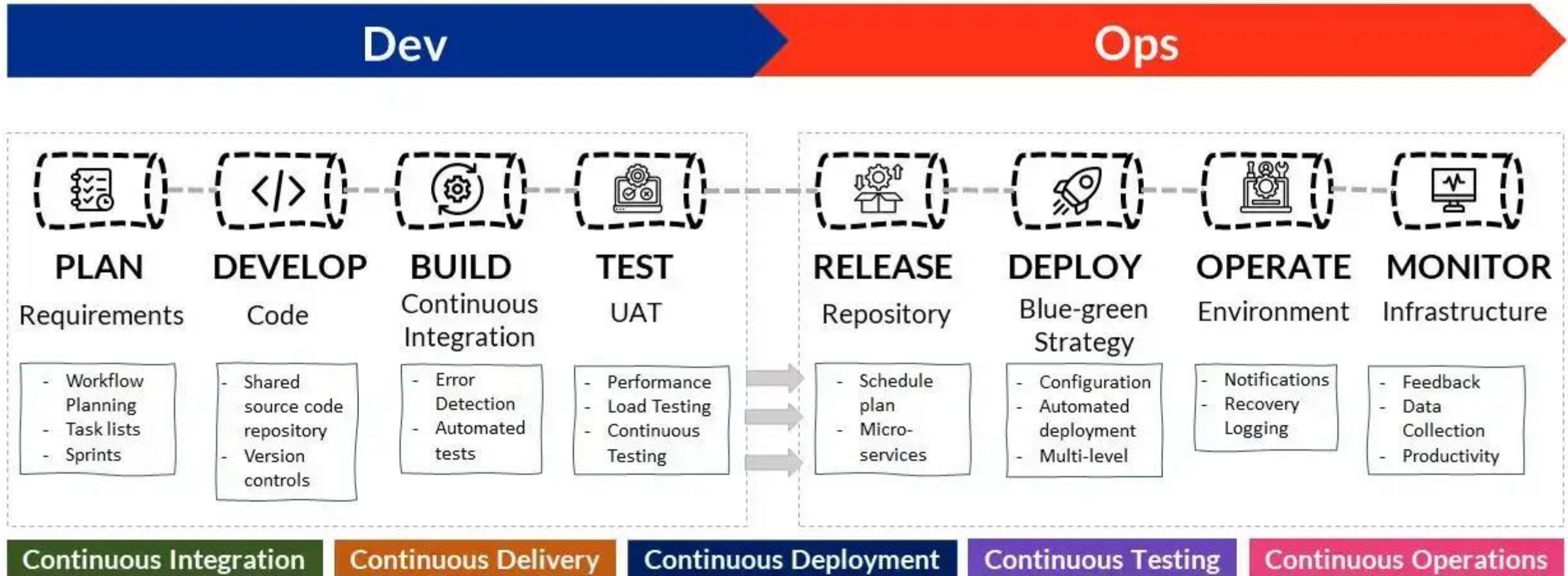
What is DevOps? A Crowded Landscape



What is DevOps?



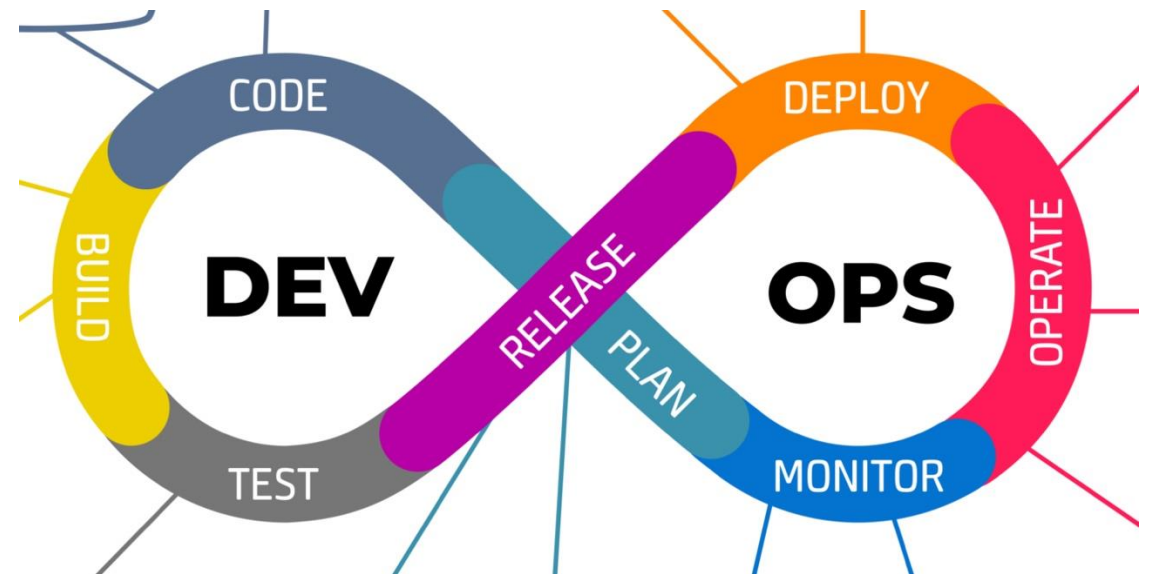
What is DevOps? A linear perspective



www.polestarllp.com

DevOps is “event-driven”

- Code Commit
- Pull Request to special branch
- Automated Testing
- Passing tests might automatically trigger CI/CD pipeline.



Headstart for Next Time

- **Install Docker.** Recommended that you install Docker locally, but you can install it on a cloud instance that you can connect to remotely.

